



Using scatterplots to understand and improve probabilistic models for text categorization and retrieval

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ARTICLE INFO

Article history:

Received 15 January 2008

Received in revised form 6 January 2009

Accepted 9 January 2009

Available online 24 January 2009

Keywords:

Naive Bayes models

Text categorization

Information Retrieval

ABSTRACT

The two-dimensional representation of documents which allows documents to be represented in a two-dimensional Cartesian plane has proved to be a valid visualization tool for Automated Text Categorization (ATC) for understanding the relationships between categories of textual documents, and to help users to visually audit the classifier and identify suspicious training data. This paper analyzes a specific use of this visualization approach in the case of the Naive Bayes (NB) model for text classification and the Binary Independence Model (BIM) for text retrieval. For text categorization, a reformulation of the equation for the decision of classification has to be written in such a way that each coordinate of a document is the sum of two addends: a variable component $P(d|c_i)$, and a constant component $P(c_i)$, the prior of the category. When plotted in the Cartesian plane according to this formulation, the documents that are constantly shifted along the x -axis and the y -axis can be seen. This effect of shifting is more or less evident according to which NB model, Bernoulli or multinomial, is chosen. For text retrieval, the same reformulation can be applied in the case of the BIM model. The visualization helps to understand the decisions that are taken to order the documents, in particular in the case of relevance feedback.

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1. Introduction

Since information has become more and more available in digital format, especially on the World Wide Web, organizing and classifying digital documents, making them accessible and presenting them in a proper way is becoming an important issue. Solutions for content organization, access and interaction are required in order to let users express their information needs, especially when the specific request is not clear in their mind [1]. Digital Library Management System (DLMS) are an example of systems that manage collections of multimedia digitalized data and include components that perform the storage, access, retrieval, and analysis of the collections of data. Information access components, such as automatic categorization of digital objects and retrieval of digital objects, allow users to interact with the system to browse, explore and retrieve resources from collections of objects. The visualization of the results returned by these components may be a key point: firstly, for system designers during the process of raw data exploration; secondly, for users to interpret results more clearly and possibly interact with them [2].

The process of transforming information into a visual form enabling the viewer to observe, browse, and understand the information is covered by the field of *information visualization*, where the graphical models are typically constructed from measured or simulated data representing objects or concepts associated with phenomena from the physical world [3]. Several techniques for visualizing multidimensional data sets have been studied [4] since, in general, there is no standard mapping into the Cartesian coordinate systems.

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Within this context, it is important to describe the structure of a component (for categorization or for retrieval) in a way that people can understand in order to promote these important components from black boxes to useful tools that convert the data into knowledge [5]. Visualization frameworks that give a straightforward graphical explanation of the organization and classification of data have been successfully designed and implemented [6]. However, these visualization approaches have rarely been applied in the specific area of textual document categorization. In fact, the representation of textual information is particularly challenging: how can the semantics of textual documents be captured and represented through graphs? [7]. For example, the framework proposed by Poulin et al. [6] allows the user to analyze the contribution of each single feature (which represents the object to be classified) to the final decision. However, this type of visual exploration seems to give the best of its functionalities when it is applied in the field of bio-informatics, or when the number of features does not exceed a limit due to the fact that they have to be displayed in a tabular-like format. In ATC, this number (which in the case of text is equal to the number of words of the vocabulary that is used to express the documents) may easily reach the tens of thousands. The same problem may be applied to other examples of visualization, in particular for those of NB classifiers [5,8].

The *Evidence Visualizer* proposed by Becker et al. [5] is a visual representation of a Simple Bayesian Classifier which helps to understand the importance of specific features in the classification process. This is done by means of pie charts which summarize the probability distribution of each feature. *Nomograms* (graphical representations of numerical relationships) are used by Mozina et al. [8] to visualize a NB classifier. Besides enabling prediction, the NB nomogram reveals the structure of the model and the relative influences of the feature values to the class probability. Both are excellent visualization approaches; however, in order to be effective, these approaches need to have no more than a few dozens of features, which is not the case for ATC.

A completely different approach which can be cited as a successful example for text classification is Self-Organizing Map (SOM). A SOM is a type of artificial neural network that is trained using unsupervised learning to produce a low-dimensional representation of the input space of the training samples [9,10]. Nevertheless, even this kind of visualization technique provides only limited forms of guidance for the user when training and testing an ATC system. Moreover, the learning rate of this type of neural network is quite slow when compared to the computational cost of the NB models.

A recent probabilistic approach on ATC that represents documents in the two-dimensional space [11] has proved to be a valid visualization tool for understanding the relationships between categories of textual documents, and helps users to visually audit the classifier and identify suspicious training data. This model defines a direct relationship between the probability of a document given a category of interest and a point in a two-dimensional space. In this light, it is possible to graph entire collections of documents in a Cartesian plane, and to design algorithms that categorize documents directly in this two-dimensional representation.

In this paper, we apply the same idea of visualization of documents of the two-dimensional representation to the case of the NB classifier and the BIM for Information Retrieval (IR). In particular, we carry out a thorough experimental analysis for NB classifiers starting from the preliminary analyses made in [12], and investigate the possibility of using the same approach for IR models such as the BIM. The experimental analysis shows that this type of visualization helps to better understand how the estimated probabilities contribute to the classifier decisions and, more importantly, how to improve classification performances significantly. Moreover, an analysis on the use of this visualization technique for the BIM show that it opens new paths for learning to rank documents and improving relevance feedback.

Section 2 introduces the basic definition for ATC and probabilistic models for ATC. Section 3 presents the two-dimensional model applied to the NB model for text categorization. Sections 4 and 5 introduce the probabilistic model for IR and the application of the two-dimensional model to it. Section 6 gives some final remarks.

2. Probabilistic models for text categorization

We refer to the problem of ATC when a general inductive process, the learner, automatically builds a text categorizer for the categories involved in the process by observing the properties of a set of pre-classified documents (this process is also known as *supervised learning*); from these properties, the inductive process learns the characteristics that a new unseen document should have in order to be categorized under a specific category [13]. ATC systems are evaluated on standard collections to make the statistical comparison of different systems valuable and possible. The actual trend of standard benchmarks for ATC witnesses the problem of the classification of news stories, for example Reuters news agencies, as a compelling task. This is not surprising since throughout the World Wide Web thousands of online newspapers using online Web sites can be accessed. For this reason, it is important to understand that user profiling can be seen as an ATC problem, where groups of users form a particular category of users interested in receiving articles of some topic, and a new user is classified under one of these profiles [14].

In the literature different approaches, such as Naive Bayes (NB) [15–17], ridge logistic regression [18] and support vector machines [19] have shown that remarkable performances can be achieved when applied to the problem of ATC.

Despite the wide use of NB models in the literature, the notation used to describe them is not standard; for this reason, we first introduce the notation used and then present the two most commonly used NB models in ATC, the Bernoulli NB model, and the multinomial NB model.

2.1. ATC definitions

Given a finite set of pre-defined categories

$$C = \{c_1, \dots, c_i, \dots, c_{|C|}\},$$

a finite set D of documents, the learning of a categorizer consists of the definition of a function for each category c_i of this kind:

$$CSV_i : D \rightarrow [0, 1],$$

where CSV is the *categorization status value*. In this case, a categorizer returns a “degree of membership” of a document d for a category c_i , that is to say, given a document $d \in D$, CSV_i returns a value for the document with respect to the category c_i . However, when the decision to assign or not d to c_i has to be taken, a threshold t may be needed such that if $CSV_i > t$, d is assigned to category c_i , otherwise it is not assigned to c_i . In the first case, the document d is said to be a *positive* example for the category c_i , i.e. d has been *accepted* for c_i ; in the second case, the document d is said to be a *negative* example for the category c_i , i.e. d has been *rejected* for c_i .

The CSV function takes on different meanings according to the learning method used: in the NB approach, the CSV_i is defined in terms of the probability that a document d belongs to the category c_i . For example, NB classifiers view $CSV_i(d)$ in terms of the conditional probability $P(c_i|d)$, that is, the probability that a document d belongs to c_i . This probability can be calculated according to Bayes’ rule as

$$P(c_i|d) = \frac{P(d|c_i)P(c_i)}{P(d)}.$$

Intuitively, $P(d|c_i)$ is the probability that a randomly drawn document from c_i is exactly d , $P(c_i)$ is the probability that a randomly drawn document belongs to c_i , and $P(d)$ is the probability that a randomly drawn document is equal to d [16].

Given a vocabulary of terms, usually indicated with $V = \{t_1, t_2, \dots, t_k, \dots, t_{|V|}\}$, where t_k is the k th term, the NB classifier assumes that all features of the examples, that is to say the terms of the documents, are independent of each other given the context of the category. This is the so-called NB assumption, and it is usually expressed with an introductory and very general formula, such as

$$P(d|c_i) = \prod P(t|c_i),$$

where the product is calculated over the terms of the vocabulary. However, it is important to remember that a “linked dependence assumption” should be considered instead of an independence assumption in order to avoid logical inconsistencies [20]. The linked dependence assumption can be expressed as the following:

$$\frac{P(d|c_i)}{P(d|\bar{c}_i)} = \frac{\prod P(t|c_i)}{\prod P(t|\bar{c}_i)},$$

where $\bar{c}_i$ is the complementary category of c_i , that is $\bar{c}_i = C - c_i$.

Having introduced the notion of complementary category $\bar{c}_i$, we need now to define the problem called *binary* categorization, also known as *one-vs-all* categorization [21]. Given a set of categories C , there are $|C|$ independent binary categorization problems where each document of D must be assigned either to category c_i or its complement $\bar{c}_i = C - c_i$. For a document d , the i th independent problem is assigning d either to category c_i or to $\bar{c}_i$. When we approach the problem in terms of binary categorization, we actually build $|C|$ binary classifiers, one for each category in C .

When a document d has to be classified under a category c_i or its complement $\bar{c}_i$, one needs to calculate if the probability $P(c_i|d)$ is greater than $P(\bar{c}_i|d)$, alternatively the odds of $P(c_i|d)$ can be used to estimate this probability. Therefore, if we assume that $P(\bar{c}_i|d) \neq 0$ (or $P(c_i|d) \neq 1$), and rewrite probabilities using Bayes’ rule, we can consider the following inequality:

$$\frac{P(c_i|d)}{1 - P(c_i|d)} = \frac{P(c_i|d)}{P(\bar{c}_i|d)} = \frac{P(d|c_i)P(c_i)}{P(d|\bar{c}_i)P(\bar{c}_i)} > 1. \quad (1)$$

If the inequality holds, we assign d to c_i , otherwise we assign it to $\bar{c}_i$.

The probability $P(c_i|d)$ is usually very close to zero. In order to avoid arithmetical errors while computing the classification decision, it is preferable to calculate the logarithm of the odds and verify this inequality

$$\log \left(\frac{P(c_i|d)}{1 - P(c_i|d)} \right) = \log \left(\frac{P(d|c_i)P(c_i)}{P(d|\bar{c}_i)P(\bar{c}_i)} \right) > 0. \quad (2)$$

2.2. Bernoulli and multinomial NB models

The Bernoulli NB model specifies that a document is represented by a vector of binary values indicating which terms occur and do not occur in the document. The number of times a term occurs in a document is not captured. When calculating

the probability of a document, the probabilities of all the values are multiplied, including the probability of non-occurrence for those terms that do not occur in the document [17].

The probability $P(d|c_i)$ is calculated as follows:

$$P(d|c_i) = \prod_{k=1}^{|V|} P(t_k|c_i)^{x_k} (1 - P(t_k|c_i))^{(1-x_k)} = \prod_{k=1}^{|V|} \left(\frac{P(t_k|c_i)}{1 - P(t_k|c_i)} \right)^{x_k} (1 - P(t_k|c_i)),$$

where x_k is a binary value that equals 1 if the term t_k is present in the document, and 0 otherwise.

The multinomial NB model specifies that a document is represented by the set of term occurrences from the document. Although the order of the terms is lost, the number of occurrences of each term in the document is captured. When calculating the probability of a document, only the probabilities of the terms that occur are multiplied [17].

The probability $P(d|c_i)$ is calculated as follows:

$$P(d|c_i) = P(|d|) \frac{|d|!}{n_{t_1}! n_{t_2}! \dots n_{t_{|V|}}!} \prod_{k=1}^{|V|} P(t_k|c_i)^{n_{t_k}},$$

where n_{t_1} indicates the number of times t_1 appeared, and the sum $\sum_{k=1}^{|V|} n_{t_k} = |d|$.

3. Two-dimensional ATC

In the two-dimensional representation of documents, two coordinates $X_i(d)$ and $Y_i(d)$ are calculated for each document d of a category c_i . In order to obtain the two coordinates in the case of the NB, we have to rewrite Eq. (2) in the following way:

$$\log(P(d|c_i)P(c_i)) > \log(P(d|\bar{c}_i)P(\bar{c}_i)),$$

which becomes

$$\underbrace{\log(P(d|c_i)) + \log(P(c_i))}_{X_i(d)} > \underbrace{\log(P(d|\bar{c}_i)) + \log(P(\bar{c}_i))}_{Y_i(d)}.$$

Each coordinate of a document is the sum of two addends: a variable component $P(d|c_i)$, the conditional probability of a document given a category, and a constant component $P(c_i)$, the prior of the category.

Fig. 1 shows the distribution of the documents of a category in the case of Bernoulli NB model, when a standard test collection of documents for ATC is used (category *corn* of Reuters-21578 10 categories dataset, described in Section 3.1). A line that shows where the decision of the classification is taken, that is to say where the inequality of Eq. (2) is verified, is plotted in both figures: below the line, the document is assigned to the category c_i , above the line, the document is assigned to $\bar{c}_i$. Documents are constantly shifted along the x -axis and the y -axis, and this effect is more evident with the Bernoulli NB model.

Thanks to this representation, we can easily understand and “see” why performances are low for particular categories. This happens when the shift along one of the axes is so substantial that the cloud of points is far from the line of the decision of classification. In such cases the classifiers consider all the documents either positive for c_i or negative.

By graphical inspection, we can already see that the decision line is not the best possible one. A translation and a rotation would result in a better separation of the two classes of points. This is the idea of the *Focused Angular Region* (FAR) algorithm

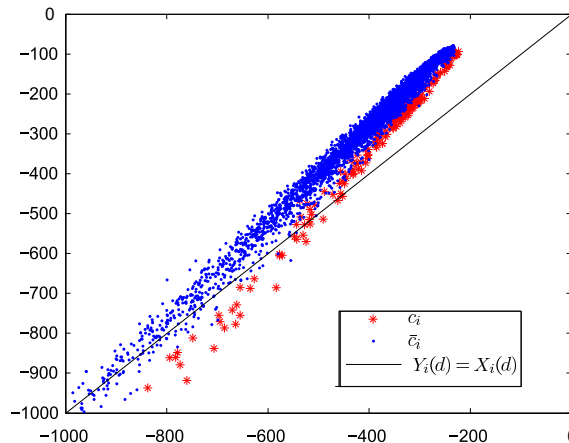


Fig. 1. NB Bernoulli model.

[22] used to improve the performances of the two-dimensional representation of documents: a better separation of the documents in c_i and $\bar{c}_i$ is reached when the best intercept $\mathcal{Q}$ together with the best angular coefficient $\mathcal{M}$ of the decision line is found. The plane is split into two regions according to the following inequality:

$$Y_i(d) < X_i(d) \cdot \mathcal{M} + \mathcal{Q}, \quad (3)$$

which means translating, with an intercept $\mathcal{Q}$, and rotating, with an angular coefficient $\mathcal{M}$, the separation line.

The general idea of the FAR algorithm may be stated as follows: let $p(0, q)$ be a point close to the origin, $|q| < \epsilon$. Let $Y_{i, pos}$ be the interpolating line of category c_i (positive documents) constrained to pass through the point $p(0, q)$ (note that the interpolating lines of positive/negative documents are found by means of standard vertical least squares fitting procedures), and let $Y_{i, neg}$ be the interpolating line of the $\bar{c}_i$ (negative documents) constrained to pass through the point $p(0, q)$. Consider the angular region whose vertex is the point $p(0, q)$, bounded by the semi-lines $Y_{i, pos}$ and $Y_{i, neg}$. Within this region the optimal separating straight line should be found, inequality 3, being the F_1 measure the cost function.

The following sections present the experimental analysis conducted on standard benchmarks for text categorization, first using the standard NB decision given by Eq. (2), then using the FAR algorithm to find a better decision of classification.

3.1. Experiments and analyses

In this section experimental results are presented. First, the datasets and evaluation measures used are discussed, and then results using standard parameters for NB models and results using the FAR algorithm are analyzed.

There are standard benchmark collections that can be used for experimental purposes for ATC. The most widely used are the Reuters datasets that collect Reuters newswire agencies that are classified under different categories. These are the Reuters-21578¹ and the Reuters RCV1.²

In order to test performances of the classifiers on collections of documents different from news agencies, we also adopted other benchmark collections more related to the World Wide Web: 20 Newsgroups,³ WebKB dataset,⁴ and 7sectors dataset.⁵ Although these three datasets have been widely used in the literature, a common split of the documents between training and test set, like in the Reuters collections, has not been decided yet. Since the proportion of training and test set can be different (e.g. 30% of the documents are used for training and 70% for test), or different random sets of documents can be built even using the same proportion between training and test sets, performances and results among experiments and classifiers are more difficult, if not impossible, to compare.

Reuters-21578: Of the 135 potential categories, we used the following subsets in the experiments: the 10 most frequent categories; the 90 most frequent categories, that is the categories with at least one training document and one test document; the 115 most frequent categories, that is the categories with at least one training document and possibly no test documents.

Reuters Collection Volume 1 (RCV1): the new Reuters corpus consists of almost 810 thousand newswires covering a whole year of Reuters news agencies, from August 20, 1996 to August 19, 1997. For the RCV1, a sample of 16 categories, which approximately follow the distribution of the documents in the whole collection [23], were chosen out of the 103 potential categories. The training set consisted of the first month of the Reuters year (20th August 1996, 19th September 1996) with 30,529 documents, while the test set consisted of the last Reuters month (20th July 1997, 19th August 1997) with 36,243 documents. A similar choice of months was made in one of the experiments presented by Damerau et al. [24].

20NewsGroups: the 20NewsGroups corpus contains approximately 20,000 newsgroup articles, equally divided into 20 categories. We used the 20-news-18828 version of this corpus, which contains 18,828 documents without duplicates; we randomly divided the documents into two sets: 50% training and 50% test.

WebKB: The WebKB data set contains Web pages gathered from university computer science departments. The pages are divided into seven categories: student, faculty, staff, course, project, department and other. The total number of documents is 8282. We used only four categories in our experiments: course, faculty, project, and student, with a total of 4199 documents, like previous experiments in the literature [25], which were randomly divided into 50% training and 50% test.

7sectors: the 7sectors database consists of 4581 HTML articles partitioned in hierarchical order. We labeled each document in this collection with its initial parent class label: basic, energy, financial, health, transportation, technology and utilities; the collection was randomly divided into two sets equally sized of training and test.

Before indexing terms, some text preprocessing was carried out on the documents of both collections: a first cleaning was made removing all the punctuation marks and converting all the letters to lower case. A standard stoplist of 571 terms⁶ was used to remove the most frequent terms of the English language; finally, the English Porter stemmer⁷ was used as the only

¹ <http://www.daviddlewis.com/resources/testcollections/reuters21578/>.

² <http://about.reuters.com/researchandstandards/corpus/>.

³ <http://people.csail.mit.edu/jrennie/20Newsgroups/>.

⁴ <http://www.cs.cmu.edu/afs/cs.cmu.edu/project/theo-20/www/data/>.

⁵ <http://www.cs.cmu.edu/afs/cs.cmu.edu/project/theo-11/www/wwwkb/index.html>.

⁶ <http://www.cs.utk.edu/cs460.is&r/labs/per1/english.stop>.

⁷ <http://www.tartarus.org/martin/PorterStemmer/>.

method of reducing the number of terms of the vocabulary. No feature selection function was used to reduce the vocabulary of terms. The Laplacian prior was used to smooth probabilities.

The effectiveness measures most widely used in ATC, inherited from IR, are defined in terms of the fraction of documents correctly and incorrectly categorized, for example: the number of *true positives* for category c_i , TP_i ; the number of *false positives* for category c_i , FP_i ; the number of *false negatives* for category c_i , FN_i ; the number of *true negatives* for category c_i , TN_i . These numbers are combined to form the two basic measures named *recall* ρ , and *precision* π :

$$\rho_i = \frac{TP_i}{TP_i + FN_i}, \quad \pi_i = \frac{TP_i}{TP_i + FP_i},$$

where the subscript i indicates that the performance is calculated on category c_i . The *F-measure* [26] was used to summarize recall and precision in a single number:

$$F_\beta = \frac{(\beta^2 + 1)\pi_i\rho_i}{\beta^2\pi_i + \rho_i}.$$

In the experiments, the F_1 measure, which means $\beta = 1$, was used to calculate the results. The *macro-averaged* measures, and the *micro-averages* measures were used to average the effectiveness of the classifier over the whole set of categories.

3.1.1. Plain vanilla NB

Table 1 summarizes the results obtained using NB models with standard parameters.

The thing that immediately stands out is that the multinomial NB model performs better than the Bernoulli model, as presented in the literature. Recall is generally higher than precision, which means that this model tends to accept documents easily. This behavior can be analyzed graphically since almost all the points are below the decision line, and all the positive points are below it (which means that they will be classified as positive for the category of interest).

The situation is almost the opposite in the case of the NB Bernoulli model. This model tends to reject documents, and considers them as a negative example for every c_i . Therefore, this model has, in general, a low recall but a higher precision. Fig. 1 confirms this; in fact, the position of the cloud of points is mainly above the decision line (which means that they will be rejected for the category of interest).

Another important thing to note is the micro-average performance of the NB Bernoulli model which is high compared to the respective macro-average performances. This can be explained in the following way: the categories where this model performs worst are those with very few documents. This is due to the fact that the probability $P(t_k|c_i)$ is practically equal to one (that is the term is present in all the documents of c_i). The result is that the shift along the x -axis is more pronounced given that the term of the coordinate $X_i(d), \sum_{k=1}^{|V|} \log(1 - P(t_k|c_i))$, contains many $\log(\sim 0)$, and therefore, a large negative number. However, since these categories contain very few documents, the contribution to the weighted average (the micro-average performance) is low and is balanced by the performance on those categories where the number of positive documents is comparable to the number of negative documents. This is also a possible explanation of why the Bernoulli model tends to perform better when documents are evenly distributed across the different categories, which is the case for the three collections 20Newsgroups, WebKB, and 7sectors.

The same does not happen to the multinomial model, where, from the graphical analysis, it seems that the particular distribution of the points is close to the origin, there is almost no horizontal or vertical shift, but “rotated” with respect to the decision line. This effect can be compensated by the FAR algorithm presented in the following section.

Table 1
Results of experiments for plain vanilla NB.

Categories	Measure	Bernoulli			Multinomial		
		ρ	π	F_1	ρ	π	F_1
Reuters-21578 (10)	Micro-average	0.7302	0.7247	0.7274	0.9383	0.8749	0.9055
	Macro-average	0.3964	0.4526	0.4226	0.8615	0.7722	0.8144
Reuters-21578 (90)	Micro-average	0.5280	0.6657	0.5889	0.8241	0.7121	0.7640
	Macro-average	0.0406	0.0542	0.0465	0.2693	0.1430	0.1868
Reuters-21578 (115)	Micro-average	0.5272	0.6651	0.5882	0.7113	0.8262	0.7645
	Macro-average	0.2490	0.2598	0.2543	0.3282	0.4337	0.3736
RCV1 (16)	Micro-average	0.7588	0.7293	0.7437	0.8880	0.8742	0.8811
	Macro-average	0.3243	0.3247	0.3245	0.5143	0.5949	0.5517
20Newsgroups	Micro-average	0.6530	0.6674	0.6601	0.7801	0.8188	0.7989
	Macro-average	0.4601	0.4899	0.4745	0.6712	0.6933	0.6821
WebKB	Micro-average	0.7888	0.8102	0.7994	0.8075	0.8398	0.8233
	Macro-average	0.6540	0.6733	0.6635	0.6819	0.6892	0.6855
7sectors	Micro-average	0.5670	0.6011	0.5836	0.6882	0.7123	0.7000
	Macro-average	0.5928	0.6404	0.6157	0.6512	0.6671	0.6591

3.1.2. NB with optimal separating line

The FAR splits the two-dimensional plane into two regions in such a way that the best possible separation of the two sets of documents, of c_i and $\bar{c}_i$, is achieved according to the F_1 measure. The plane is split into two regions according to the inequality 3.

Table 2 gives an overview of the results obtained using the FAR algorithm to find the optimal decision line.

Performances improve significantly for every measure and average for the Bernoulli NB model, and it reaches the performance of the Multinomial model of Table 1. This is an important achievement since it is possible to show experimentally that the difference in performance of the two models is not due to the different assumptions of how documents are generated (binomial or multinomial distribution) but to the different position that the clouds of points have in the space (which is given by how probabilities are calculated). An adjustment of the decision line is sufficient to make the Bernoulli model as powerful, in terms of classification performances, as the multinomial one. A different approach would be to estimate probabilities with a different formula, as shown by Eyheramendy et al. [15].

The low performances of the multinomial model, compared to the “plain vanilla” experiments, led to the discussion on how the FAR algorithm calculates the decision line, and how the graphical inspection can help to understand anomalies, which is discussed in the following section.

3.2. Further considerations

Compared to the studies in the literature [27,18], the two-dimensional approach for the NB models achieves the same performances of the best results that make use of feature selection functions, or when different probability smoothing approaches are tested [15]. This is an important finding because it shows that there is no need to reduce the dimension of the vocabulary, with a consequent loss of variety of terms, or to tune parameters to adjust probabilities, which possibly leads to arithmetical inconsistencies (for example total probabilities that do not sum to one), in order to get the best performances out of NB models.

The graphical representation not only gives indications on the classification decisions and helps to understand the NB models, but also gives suggestions on how to improve performances and correct the decisions. A practical example is shown in Fig. 2: this figure which depicts category *cotton* clearly shows where a first implementation of the FAR algorithm presented some problems. In particular, the dashed line shows the decision of the first implementation of the classifier, and it is immediately visible that the algorithm is not working as expected. This analysis would not have been possible looking at the numbers of the performances since they would have indicated only a low number that might have indicated a particularly difficult category. Indeed, thanks to the graphical analysis a bug in the update of the algorithm that finds parameters $\mathcal{Q}$ and $\mathcal{M}$ was spotted.

It is also possible to study the effect of the selection of a subset of terms of the vocabulary which represents a document. Fig. 3 shows a category represented by only 200 terms. Fig. 4 shows the same category represented by only 1000 terms. It is possible to see the shift and rotation of the clouds of points with respect to the other figures, and also the difference of scale of the axes. Even in these cases, an adjustment of the decision line would bring benefits to the classification performances.

Finally, a more general way to write Eq. (2) is to have a threshold q instead of a zero in the right-hand side of the inequality. This means having in the two-dimensional representation, and in Eq. (3), $\mathcal{M} = 1$ and $\mathcal{Q} = q$. Therefore, using a threshold in a “classical” way means shifting the line that separates the points of c_i and $\bar{c}_i$ parallel to the bisecting line of the first and third quadrant of the two-dimensional plane.

Table 2

Results of experiments for NB using FAR algorithm.

Categories	Measure	Bernoulli			Multinomial		
		ρ	π	F_1	ρ	π	F_1
Reuters-21578 (10)	Micro-average	0.9275	0.8810	0.9037	0.9110	0.9130	0.9120
	Macro-average	0.8682	0.7715	0.8170	0.8195	0.8122	0.8158
Reuters-21578 (90)	Micro-average	0.7551	0.7535	0.7543	0.7193	0.4031	0.5167
	Macro-average	0.2479	0.3551	0.2920	0.2650	0.2675	0.2662
Reuters-21578 (115)	Micro-average	0.7575	0.7515	0.7545	0.7196	0.2561	0.3778
	Macro-average	0.4107	0.4845	0.4446	0.4269	0.2208	0.2910
RCV1 (16)	Micro-average	0.8452	0.7517	0.7957	0.8462	0.9097	0.8768
	Macro-average	0.5564	0.4393	0.4909	0.5098	0.6202	0.5596
20Newsgroups	Micro-average	0.7429	0.7677	0.7551	0.7701	0.8123	0.7906
	Macro-average	0.5723	0.6198	0.5951	0.6534	0.6845	0.6686
WebKB	Micro-average	0.7920	0.8231	0.8073	0.8123	0.8297	0.8209
	Macro-average	0.6912	0.6967	0.6635	0.6702	0.6624	0.6663
7sectors	Micro-average	0.6546	0.6589	0.6587	0.6745	0.6900	0.6822
	Macro-average	0.6230	0.6412	0.6320	0.6519	0.6598	0.6558

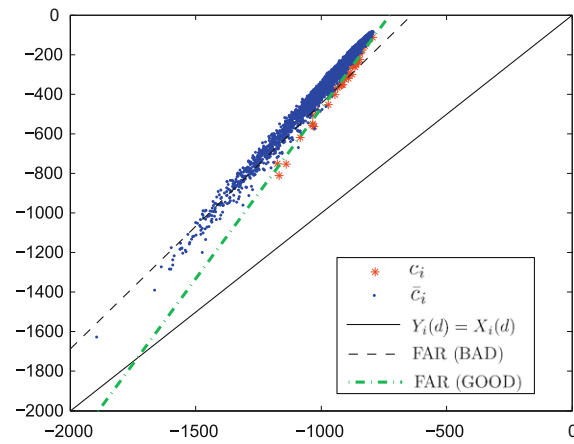


Fig. 2. NB Bernoulli model. Optimizing FAR algorithm.

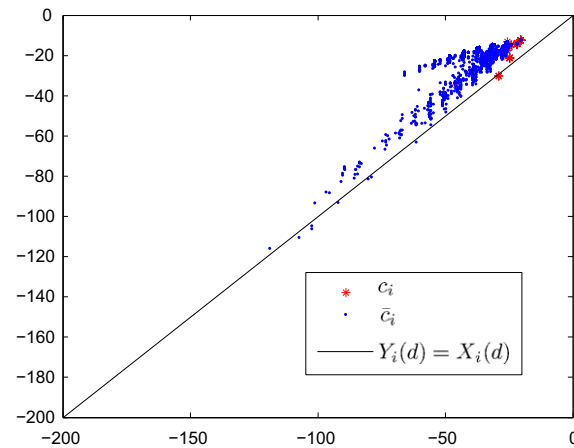


Fig. 3. NB Bernoulli model, *interest* category, 200 features (Reuters-21578 10 categories).

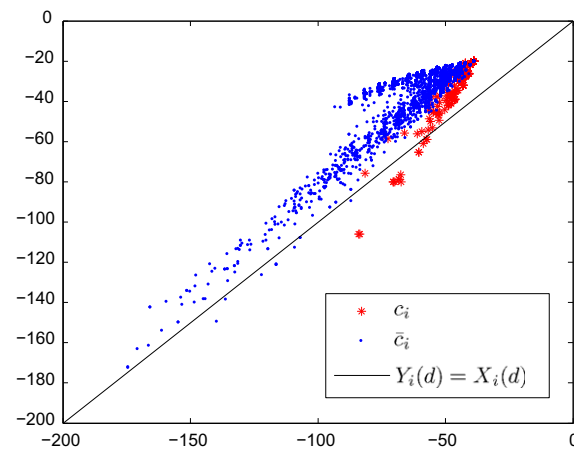


Fig. 4. NB Bernoulli model, *interest* category, 1000 features (Reuters-21578 10 categories).

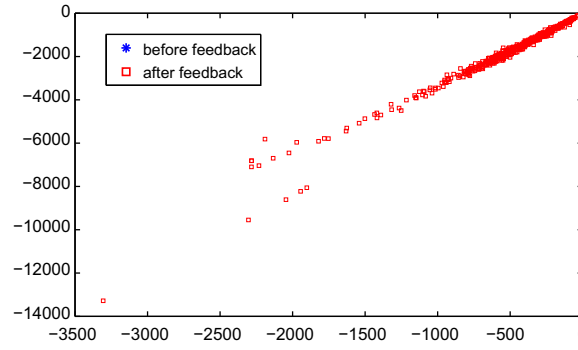


Fig. 5. Examples of the use of relevance feedback using five documents for blind feedback.

4. Probabilistic models for Information Retrieval

Retrieving documents is something similar to classifying them into “relevant” (indicated with R) documents and “non-relevant” (indicated with $\bar{R}$) documents.

Among the different probabilistic models in Information Retrieval [28] we decided to analyze one of the simple, but powerful, models: the BIM [29]. The BIM represents documents by means of “binary” vectors in the same fashion as the Bernoulli model for text categorization presented in Section 2.2. However, in this case we have a query q and, for each document d , we need to compute $P(R|q, d)$ which is the probability of relevance given the document and the query. The basic assumption is that terms are distributed differently within relevant and non-relevant documents. In the BIM, the documents and queries are both represented as binary term vectors and we can derive the same equations presented in Section 2.2 having relevant or non-relevant instead of c_i or $\bar{c}_i$.

Since what matters in retrieving documents is the ranking of documents, it is possible to rank documents by their log-odds of relevance by means of the following sum:

$$\log \left(\frac{P(R|q, d)}{P(\bar{R}|q, d)} \right) = \sum_{k=1}^{|V|} x_k \log \left(\frac{P(t_k|R, q)(1 - P(t_k|\bar{R}, q))}{P(t_k|\bar{R}, q)(1 - P(t_k|R, q))} \right), \quad (4)$$

where x_k in this case is a binary value that equals 1 if the term t_k is present in the document and in the query, and 0 otherwise. The logarithm is known in literature as the *relevance weighting formula*.

5. Two-dimensional retrieval

Following the same reasoning as the two-dimensional representation of documents for ATC, in order to obtain the two coordinates for the BIM we can rewrite the log-odds:

$$\underbrace{\sum_{k=1}^{|V|} x_k \log \left(\frac{P(t_k|R, q)}{(1 - P(t_k|R, q))} \right)}_{X(d)} > \underbrace{\sum_{k=1}^{|V|} x_k \log \left(\frac{P(t_k|\bar{R}, q)}{(1 - P(t_k|\bar{R}, q))} \right)}_{Y(d)}.$$

If the two sets of relevant and non-relevant documents were known, we could draw the points in the two-dimensional space in the same way we did for ATC. However, this information is not known a priori; therefore, it is usual to start from an estimate where all documents are non-relevant, apply some feedback on the list of documents thus calculated and re-estimate probabilities of documents to obtain a different ranking. Our preliminary analysis on the use of the visualization technique focuses on the application of “pseudo-relevance feedback”, which means selecting the documents which are ranked at the top of the list.

Fig. 5 shows the visualization of the collection of documents before and after relevance feedback using the Los Angeles Times news collection, on the topic “What brands are marketed by Nestle around the world?” is considered, which was the first topic available for the Cross-Language Evaluation Forum⁸ (CLEF). The points are placed along a straight line, all with the same abscissa. This is correct since all the documents have the same probability of being relevant (equal to zero), and a different probability of being non-relevant. The blind-relevance feedback step consists in selecting those documents which are less non-relevant, which means take the documents with the lowest $Y(d)$.

The squares in the figure show the collection of documents displayed after recalculation of probabilities. The five documents selected for feedback are clearly separated from the remainder of the collection. The effect is even more evident when new documents are added to the set of relevant documents.

⁸ <http://www.clef-campaign.org>.

Table 3

Results of experiments for BIM model using blind relevance feedback, and using the two-dimensional view.

Measure	BIM	BIM plus feedback	2D BIM	2D BIM plus feedback
Mean average precision	0.2015	0.2614	0.2101	0.2705

The final aim would be that of adding documents until the two clouds of points stabilize indicating that no further improvement in the separation of relevant and non-relevant documents can be achieved.

5.1. Experiments and analysis

The collection used for the experiments is the English news collection of the Los Angeles Times 2002 which consists of 113,005 documents made available by the CLEF. The standard 50 topics were used. Documents and topics were indexed using only a stoplist and no stemming. We indexed the text of the documents contained in the tags <HD> (headline) and <TX> (text), while we indexed the text of the topics contained in the <TITLE> (title) and <DESC> (description) fields. Blind relevance feedback was used to select the top ten ranked documents and re-order the list of documents.

5.1.1. Plain vanilla BIM

Two different experiments were carried out. The first experiment was performed using the simple BIM, while the second experiment was carried out using the top ten documents to perform relevance feedback and estimate the probability of the terms in the relevant documents. Results are summarized in Table 3.

The first experiment achieved a mean average precision over the 50 topics around 20%, while the experiment that used the blind relevance feedback achieved a mean average precision over the 50 topics around 26%. If we compare these numbers with the best experiments of CLEF 2007 [30], we can see a significant difference. However, this is reasonable since we are performing experiments with the simplest probabilistic model without any additional tool. The aim of this experimental analysis is in fact to understand if the two-dimensional model can affect the performance in some way.

5.1.2. Two-dimensional BIM

For the two-dimensional experiments we tried a different approach. Instead of ordering the documents according to the relevance score of Eq. (4), we ordered the list of documents first according to the x -axis and then according to the y -axis. For example if d_1 and d_2 are two documents, we rank d_1 higher than d_2 in the list if $x_1 \geq x_2$ (higher probability of being relevant) and $y_1 \leq y_2$ (lower probability of being non-relevant). In this way, documents with the same score can be ranked differently if the x and y coordinates are different.

The first experiments without feedback performed in a similar way to the first experiment of the BIM, with a mean average precision around 21%. The second experiments, with blind relevance feedback, achieved a mean average precision around 27%.

The improvement using this type of ordering is not significant with respect to the normal BIM. However, there is still some margin for improvements if the FAR algorithm is used even in this case. This type of analysis is still underway, but there are promising results confirmed by Fig. 5 where the possibility of better separating the relevant and non-relevant documents is clear.

6. Conclusions

The approach presented in this paper introduces a possible use of the two-dimensional representation of documents which allows documents to be represented in a two-dimensional Cartesian plane and which has proved to be a valid visualization tool for ATC. This visualization tool is used to understand the relationships between categories of textual documents, and to help users to visually audit the classifier and identify suspicious training data. In order to obtain the two coordinates in the case of the NB, a reformulation of the equation for the decision of classification has to be written in such a way that each coordinate of a document is the sum of two addends: a variable component $P(d|c_i)$, and a constant component $P(c_i)$, the prior of the category. When plotted on the Cartesian plane according to this formulation, the documents that are constantly shifted along the x -axis and the y -axis can be seen. This effect of shifting is more (or less) evident according to which NB model, Bernoulli or multinomial, is chosen. The same reformulation has been applied in the case of the BIM model for Information Retrieval.

Experimental results on standard benchmark collections confirm that the idea of representing probabilities of NB in the two-dimensional space and finding a separating line different from the one expected using Eq. (2) is valid. Moreover, the behavior of classifiers in particular situations can be understood and explained by means of a graphical inspection. Experiments on collections of different domains was useful for understanding the behavior of the classifiers with and without the use of the FAR algorithm. In particular, the multinomial model performs better in all the tests, in particular in the case of news agencies; however, the difference in performance between the multinomial and the Bernoulli model significantly decreases when the FAR algorithm is used. A possible explanation for this result is that the Bernoulli model tends to perform

better when documents are evenly distributed across the different categories, which is the case for the three collections 20Newsgroups, WebKB, and 7sectors. Finding a better separating line in a situation where the number of documents in each category is unbalanced helps the Bernoulli model to contrast the effect of the shift along the x-axis given by the sum of the vocabulary of terms. It is important to stress that it was possible to understand this particular effect only with the use of the two-dimensional visualization.

Preliminary results on the possible use of this visualization technique for IR are encouraging; in particular, in the case of the use of blind relevance feedback for the BIM we saw that it is possible to display the quality of the feedback in terms of separation between the set of relevant and non-relevant documents.

Current work on the joint use of NB models and the two-dimensional representation is investigating the possibility of drawing the cloud of points incrementally, making the visualization of the contribution of each term possible. This would help in understanding which terms better discriminate the two sets of points.

There are also large margins for improvements regarding the interaction of the user with the system. The study of HCI techniques to interact with the system and directly change the estimated probability is also under investigation. A user may enter the value of preference of the terms that he considers more valuable and interesting of a particular category and receive visual feedback on the screen to understand if the change of this value gives a better or a worse separation of the categories. The user may also select the relevant documents for some topics and see the clouds of relevant and non-relevant documents shifting through the space.

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